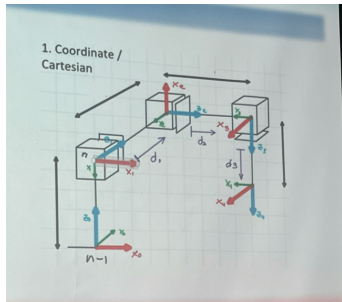
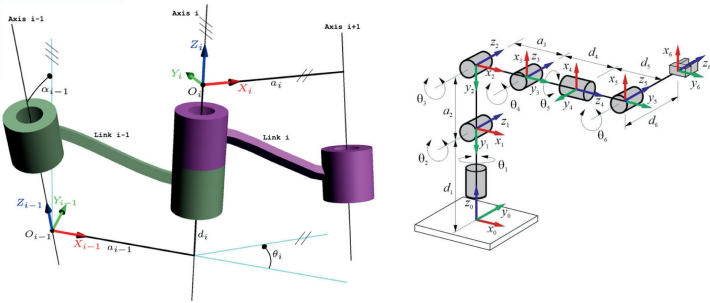


## Setup coordinates

### Common Rules for Frame Assignment (การกำหนดระบบแกนในหุ่นยนต์)

- กำหนด GCS
- กำหนด LCS (ระบบแกนที่  $i$ ) ของข้อต่อถัดไปโดยแกน  $z$  เชื่อมโยงกับการเคลื่อนที่ของข้อต่อ
  - Prismatic:** กำหนดให้แกน  $z$  อยู่บนแนวทางการเคลื่อนที่เชิงเส้นของข้อต่อ
  - Revolute/Rotary:** กำหนดให้แกน  $z$  อยู่บนแนวการหมุนของข้อต่อ
- กำหนด  $x$ -axis is perpendicular to both  $z_{i-1}$  and  $z_i$  ( $x_i = z_i \times z_{i-1}$ )
- กำหนดจุดกำเนิดแกน LCS  $i$  โดยให้อยู่ที่จุดตัดของแกน  $z_i$  และ  $x_i$



## Take derivative of the translation matrix to find the jacobian

### Relative motion

$$\vec{r}_{B/A} = \vec{r}_B - \vec{r}_A = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\vec{v}_{B/A} = \frac{d\vec{r}_{B/A}}{dt} = \frac{dx(t)}{dt}\hat{i} + \frac{dy(t)}{dt}\hat{j} + \frac{dz(t)}{dt}\hat{k}$$

$$\vec{a}_{B/A} = \frac{d^2x(t)}{dt^2}\hat{i} + \frac{d^2y(t)}{dt^2}\hat{j} + \frac{d^2z(t)}{dt^2}\hat{k}$$

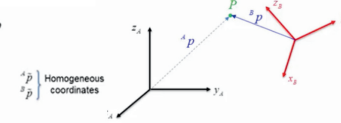
## Write the transformation matrix

### Recall Position of RR

- Point  $P$  of frame  $\{B\}$  in frame  $\{A\}$ :  ${}^A P = {}^A T_B + {}^A R_B {}^B P$
- In a more compact way:

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A T_B \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

Homogeneous transformation matrix



$$\begin{bmatrix} x_0 \\ y_0 \\ z_0 \\ 1 \end{bmatrix} = T_n^0 = \begin{bmatrix} x_n \\ y_n \\ z_n \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & d_1 \\ r_{21} & r_{22} & r_{23} & d_2 \\ r_{31} & r_{32} & r_{33} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_n \\ y_n \\ z_n \\ 1 \end{bmatrix}$$

ข้อ.หุ่น RR,  $T_2^0 = \begin{bmatrix} c(\theta_1 + \theta_2) & -s(\theta_1 + \theta_2) & 0 & l_1 c(\theta_1) + l_2 c(\theta_1 + \theta_2) \\ s(\theta_1 + \theta_2) & c(\theta_1 + \theta_2) & 0 & l_1 s(\theta_1) + l_2 s(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$T = \begin{bmatrix} \text{Rot } 3 \times 3 & \text{Trans } 1 \times 3 \\ r_{11} & r_{12} & r_{13} & d_1 \\ r_{21} & r_{22} & r_{23} & d_2 \\ r_{31} & r_{32} & r_{33} & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

### Velocity of End effector (RR Robot)

กรณีหุ่น RR,  $d_3 = 0$  และมี 2 ข้อต่อ จึงได้

$$\frac{dX_E}{dt} = \frac{df(\theta_1(t), \theta_2(t))}{dt} = \frac{\partial f(\theta_1, \theta_2)}{\partial \theta_1} \frac{d\theta_1}{dt} + \frac{\partial f(\theta_1, \theta_2)}{\partial \theta_2} \frac{d\theta_2}{dt}$$

$$\frac{dY_E}{dt} = \frac{dg(\theta_1(t), \theta_2(t))}{dt} = \frac{\partial g(\theta_1, \theta_2)}{\partial \theta_1} \frac{d\theta_1}{dt} + \frac{\partial g(\theta_1, \theta_2)}{\partial \theta_2} \frac{d\theta_2}{dt}$$

$$\begin{bmatrix} X_E \\ Y_E \end{bmatrix} = \begin{bmatrix} L_1 c(\theta_1) + L_2 c(\theta_1 + \theta_2) \\ L_1 s(\theta_1) + L_2 s(\theta_1 + \theta_2) \end{bmatrix}$$

$$\begin{bmatrix} \dot{X}_E \\ \dot{Y}_E \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial \theta_1} & \frac{\partial f}{\partial \theta_2} \\ \frac{\partial g}{\partial \theta_1} & \frac{\partial g}{\partial \theta_2} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = \frac{\partial f(\theta)}{\partial \theta} \dot{\theta}$$

Matrix form

Jacobian Matrix

$$\begin{bmatrix} -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

